## Experiment 27: Logistic Regression for Customer Churn Prediction

**Aim:** To predict whether a customer will leave a company using Logistic Regression.

**Algorithm:**

1. Import Logistic Regression.
2. Create customer training data.
3. Train the model.
4. Get customer details from the user.
5. Predict churn status.

**Code:**

```
from sklearn.linear_model import LogisticRegression

X = [[10, 1], [20, 2], [30, 3],
    [80, 12], [90, 15], [100, 18]]

y = [1, 1, 1, 0, 0, 0]

model = LogisticRegression()
model.fit(X, y)

usage = int(input("Enter usage minutes: "))
duration = int(input("Enter contract duration: "))

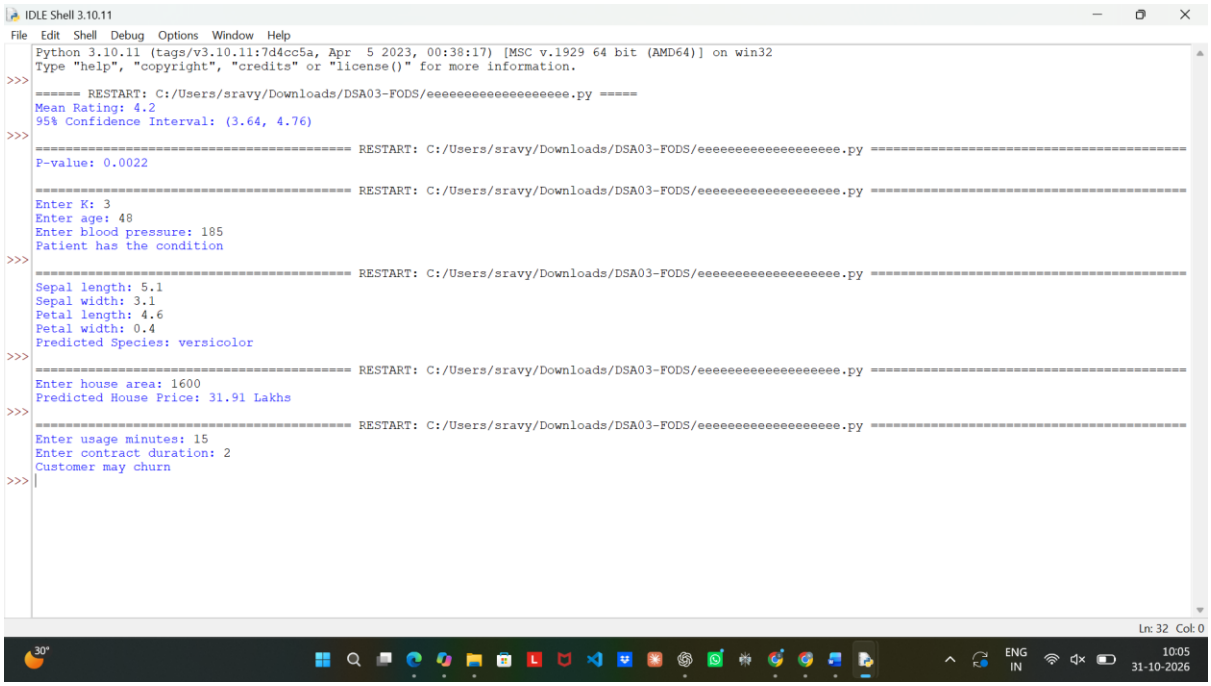
result = model.predict([[usage, duration]])

if result[0] == 1:
   print("Customer may churn")
else:
   print("Customer may not churn")
```

**Input:**

```
Enter usage minutes: 15
```

Enter contract duration: 2




**Output:**

Customer may churn

**Result:** Customer churn was successfully predicted using Logistic Regression.